

Auteur: Thijs Van Schel
 Studierichting: Informatica
 Oefeningenreeks 4A nr. 60.14

$$\begin{aligned}
 & \int \frac{6x + 2 - \sqrt{4x + 1}}{2x + 1 + \sqrt{4x + 1}} dx \\
 & u = \sqrt{4x + 1} \Rightarrow x = \frac{u^2 - 1}{4} \Rightarrow dx = \frac{u}{2} du \\
 & = \int \frac{\frac{3u^2 + 1}{2} - u}{\frac{u^2 + 1}{2} + u} \cdot \frac{u}{2} du \\
 & = \int \frac{\frac{3u^2 - 2u + 1}{2}}{\frac{u^2 + 2u + 1}{2}} \cdot \frac{u}{2} du \\
 & = \int \frac{3u^2 - 2u + 1}{u^2 + 2u + 1} \cdot \frac{u}{2} du = \frac{1}{2} \int \frac{3u^3 - 2u^2 + u}{(u + 1)^2} du \\
 & u = t + 1 \Rightarrow du = dt \\
 & = \frac{1}{2} \int \frac{3(u - 1)^3 - 2(u - 1)^2 + (u - 1)}{u^2} du \\
 & = \frac{1}{2} \int \frac{3(u^3 - 3u^2 + 3u - 1) - 2(u^2 - 2u + 1) + u - 1}{u^2} du \\
 & = \frac{1}{2} \int \frac{3u^3 - 11u^2 + 14u - 6}{u^2} du \\
 & = \frac{1}{2} \int \left(3u - 11 + \frac{14}{u} - \frac{6}{u^2} \right) du \\
 & = \frac{1}{2} \left(\frac{3u^2}{2} - 11u + 14 \ln |u| + \frac{6}{u} \right) + C \\
 & = \frac{3(t + 1)^2}{4} - \frac{11(t + 1)}{2} + 7 \ln |u + 1| + \frac{3}{u + 1} + C \\
 & = \frac{3(\sqrt{4x + 1} + 1)^2}{4} - \frac{11(\sqrt{4x + 1} + 1)}{2} + 7 \ln |\sqrt{4x + 1} + 1| + \frac{3}{\sqrt{4x + 1} + 1} + C
 \end{aligned}$$

References